



Alberta Public Health Disease Management Guidelines

Severe Acute Respiratory Infection (SARI)



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Public and Rural Health Division

Alberta Health

Severe Acute Respiratory Infection | Alberta Health, Government of Alberta

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Revisions

Revision Date	Document Section	Description of Revision
August 2024	General	Entire guideline revised to align with current information.
	Case Definition	- Updated case definition. - Added subsection with exposure criteria.
	Reporting Requirements	- Added information on reclassification and reporting if a causative organism is identified with or beyond 72 hours. - Updated reporting timelines. - Updated link to national case reporting form from the Public Health Agency of Canada. - Added subsection on International Health Regulation.
	Clinical Assessment and Epidemiology	New section.
	Public Health Management	New section.

Case Definition

(Previously Severe Respiratory Illness [SRI])

Confirmed Case

A person admitted to hospital with:

- Respiratory symptoms i.e. new onset (or exacerbation of) chronic cough or breathing difficulty, together with fever^(A) (over 38° Celsius)

AND

- Evidence of severe illness progression, i.e.,:
 - Radiographic evidence of infiltrates consistent with pneumonia, or a diagnosis of acute respiratory distress syndrome (ARDS)^(B) or severe influenza-like illness (ILI)^(C), which may also include complications such as encephalitis, myocarditis or other severe and life-threatening complications

AND

- Either admission to the ICU, or another area of the hospital where critically ill patients are cared for, **OR** where mechanical ventilation is required

AND

- No alternate diagnosis within the first 72 hours of hospitalization, i.e.:
 - Results of preliminary clinical and/or laboratory investigations within the first 72 hours of hospitalization cannot ascertain a diagnosis that reasonably explains the illness (i.e., have ruled out presence of a known viral or bacterial respiratory pathogen)

AND

- Who meets one or more of the [exposure criteria](#).

OR

A deceased person with:

- History of unexplained severe acute respiratory illness, including fever and new onset of (or exacerbation of chronic) cough or breathing difficulty, resulting in death

AND

- Autopsy findings consistent with the pathology of ARDS^(B) without an identifiable cause (i.e., a pathogen)

AND

- No alternate diagnosis that reasonably explains the illness

AND

- Who meets one or more of the [exposure criteria](#).

^(A)Fever may not be prominent in patients under 5 years or 65 years and older as well as in immunosuppressed individuals. Failure to take temperature should not rule out a history of self-reported fever. Clinical judgement should always prevail regarding these groups.

^(B) Documentation of evidence of ARDS should be noted on either the radiologist's or medical examiner's report.

^(C) ILI includes the acute onset of respiratory illness which includes cough (new or worsening chronic cough) and one or more of the following symptoms: fever, shortness of breath, sore throat, myalgia, arthralgia, and/or prostration. Note, GI symptoms may also be present in children under five years of age.

Exposure Criteria

A person with one or more of the following⁽¹⁾:

- Residence in, or recent travel (within 10 days of illness onset) to, an area (including other areas of Canada or Alberta) where novel influenza virus or other emerging/re-emerging pathogen(s) have recently been detected or are known to be circulating in humans and/or animals^(D).

OR

- Close contact^(E) with an ill person (i.e., experiencing respiratory symptoms) who resided in or recently returned from an area where cases of novel influenza virus or other emerging/re-emerging pathogen(s) have recently been detected or are known to be circulating in humans and/or animals, within 10 days prior to onset of symptoms.

OR

- Settings in which there have been mass die-offs or illness in domestic poultry or swine in the previous six weeks.

OR

- Occupational exposure involving direct health care, laboratory, or animal exposure (within 10 days of illness onset):
 - Health care workers (HCWs)^(F) who work in an environment where patients with SARI are being cared for, particularly patients requiring intensive care;
 - Laboratory exposure in a person who works directly with biological specimens from individuals with acute respiratory infections;
 - Working directly with potentially infected birds/pigs/other animals:
 - Poultry/swine farm,
 - Poultry/swine processing plant,
 - Culling Poultry/swine (catching, bagging, transporting, or disposing of dead birds/swine),
 - Wild bird/animal rescue/rehabilitation,
 - Live bird/animal market,
 - Pet bird/animal dealer or trader,
 - Chef working with live or recently killed domestic poultry, swine, or other potentially affected animals,
 - Veterinary work,
 - Public health inspector/regulator.

^(D) Refer to the World Health Organization Human Animal Interface for the most recent information
http://www.who.int/influenza/human_animal_interface/en/

^(E) Close contact is defined as: anyone who provided care for the patient, including a health care worker or family member, or who had other similarly close physical contact; anyone who stayed at the same place (e.g., lived with, visited) as a confirmed case while the case was symptomatic.

^(F) HCWs include all health practitioners and all individuals at increased risk for exposure to, and/or transmission of, a communicable disease because they work, study, or volunteer in one or more of the following health care environments: hospital, nursing home (facility living), supportive living accommodations, or home care setting, mental health facility, community setting, office or clinic of a health practitioner, clinical laboratory.

Reporting Requirements

Reported SARI cases, by definition, will not have an organism identified in the first 72 hours of hospitalization. If a causative organism is identified, either within or beyond 72 hours, reclassification and reporting are required as follows:

- a notifiable disease – as per disease-specific reporting requirements, OR
- an emerging disease – as per reporting requirements for “Diseases in rare and unusual form” or “Section 15 of the Alberta *Public Health Act*,” OR
- not notifiable – if it does not meet reporting requirements.

Physicians

- Physicians shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed cases of severe acute respiratory infection (SARI) in the prescribed form by the fastest means possible (FMP).

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- The MOH (or designate) of the zone where the case currently resides shall notify the Chief Medical Officer of Health (CMOH) (or designate) by FMP of all confirmed cases.
- The MOH (or designate) of the zone where the case currently resides shall forward the initial [Emerging Respiratory Pathogens and Severe Acute Respiratory Infection \(SARI\) Case Report Form](#) of all confirmed cases to the CMOH (or designate) within 24 hours of notification and the final form (amendment) within two weeks of notification.
- For out-of-province and out-of-country cases and/or contacts, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report (if applicable), and
 - other relevant clinical/epidemiological information.

International Health Regulation Reporting

Under the *International Health Regulations* (2005), enhanced reporting of SARI to the World Health Organization (WHO) is mandatory. Alberta Health notifies the Public Health Agency of Canada (PHAC) of every confirmed case of SARI and PHAC notifies the WHO within 24 hours of initial notification.

Clinical Assessment and Epidemiology

Etiology

SARI has no recognized etiology and is reportable when no causative agent is identified within 72 hours of hospitalization.

Clinical Presentation

Clinical presentation of SARI can vary, including pneumonia or acute respiratory distress syndrome (ARDS), a rapid onset of widespread inflammation in the lungs culminating in pulmonary edema.⁽²⁾ Patients with SARI generally require mechanical ventilation.⁽³⁾

Diagnosis

The diagnosis of SARI can be challenging. It is a clinical illness without a confirmed pathogen identified within the first 72 hours of presentation to hospital. Consultation with an Infectious Disease Specialist and the Virologist-on-call is recommended. Consideration should be given to ensuring conventional bacteria (e.g., *Mycoplasma pneumoniae*, *Legionella pneumophila*) and common respiratory illnesses (e.g., SARS-CoV-2, seasonal influenza, parainfluenza, respiratory syncytial virus, adenovirus, human metapneumovirus, rhinovirus/enterovirus) are ruled out prior to considering more unusual etiologies (e.g., MERS/SARS/novel coronavirus, and avian influenza)⁽³⁾.

For more information, refer to the [Protocol for Microbiological Investigations of Severe Acute Respiratory Infections \(SARI\)](#).

Treatment

Decisions on treatment protocols will depend on the suspected pathogen(s), severity of illness, and underlying conditions. Respiratory support should be guided by a clinician with expertise in critical care with additional treatment determined in consultation with an infectious disease specialist, as required.

Reservoir

Unknown.

Transmission

Unknown.

Incubation Period

Unknown.

⁽³⁾ Use of procedures such as non-invasive ventilation or providing nebulized medications are strongly discouraged in cases of SARI or new and emerging respiratory pathogens.^(5,6) To prevent nosocomial transmission, these procedures require attention to IPC practices and the use of negative pressure rooms.

Period of Communicability

Unknown.

Host Susceptibility

Unknown. Generally, people with occupational or travel exposure are at greater risk of acquiring SARI.⁽⁴⁾ The elderly, young children and immunocompromised are more likely to have severe illness.

Incidence

SARI is nationally notifiable; however, published surveillance data on national incidence is not available.

SARI, previously reported as Severe Respiratory Illness (SRI), has been notifiable in Alberta since 2011^(H). Three cases were reported between 2011 and 2022: 2014 (1) and 2015 (2); two of them were fatal. For more information on reported cases in Alberta, refer to the [Interactive Health Data Application \(IHDA\)](#).

^(H) One case of SRI was reported in 2009 as part of surveillance during the H1N1 pandemic.

Public Health Management

Key Investigation

- Confirm that the individual meets case definition.
- Obtain a history of illness including the date of onset and signs and symptoms.
- Investigate potential source(s) of infection through a thorough travel and exposure history (see [Exposure Criteria](#)).
- Identify close contacts⁽¹⁾ of the case.
- In addition, identify contacts that may have been exposed to the same suspected source(s) of infection as the case.

NOTE: If a known notifiable disease is subsequently identified, refer to the appropriate [Alberta Public Health Disease Management Guideline](#) for specific guidance on public health follow-up.

Management of a Case

- Individuals with SARI should be isolated in accordance with hospital infection prevention and control guidelines.
- Consultation with an infectious disease specialist is recommended.

Management of Contacts

- In consultation with the CMOH, the MOH may consider additional interventions for close contacts and/or individuals exposed to a common suspected source of infection (e.g., poultry/swine), depending on the suspected or confirmed [exposure criteria](#). This may include, but is not limited to, symptom assessment, testing, quarantine and/or exclusion.

Preventive Measures

- Provide education on the following for the general public⁽⁴⁾:
 - For anyone planning to travel outside of Canada, visit the Government of Canada website for [Travel Advice and Advisories by destination](#) to determine if there are any health precautions they should take for their particular destination,
 - Avoid contact with sick or dead animals or birds,
 - If contact is unavoidable, use disposable gloves and wash hands before and after contact, and utilize respiratory protection (i.e., mask) if possible.
 - Practice good hand hygiene and respiratory/cough etiquette,
 - Follow good food safety and food hygiene practices, and
 - Adhere to recommended infection prevention and control practices in health care facilities.
- Other Government of Canada Resources:
 - [Wild birds and avian influenza – handling guidelines](#)
 - [Advice for veterinarians and swine producers – swine influenza](#)

⁽¹⁾ Close contact is defined as: anyone who provided care for the patient, including a health care worker or family member, or who had other similarly close physical contact with anyone who stayed at the same place (e.g., lived with, visited) as a probable or confirmed case while the case was symptomatic.

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